

Homework 03

Question 1: Build an explain-to-me skill

I created `explain-to-me/SKILL.md`, installed it, and explicitly tested it on Question 2.

Question 2: Ridge regression with nearly collinear predictors

Answer (a)

The computed correlation is **0.99770569**, and the largest and smallest singular values are **14.141480** and **0.4753575**, respectively. The first two columns are almost identical. Moving their coefficients in opposite directions therefore changes the fitted response very little, so noise can cause large changes in their estimated difference. Their sum acts along a strongly represented direction and is much more precisely estimated. The small singular value identifies a weakly represented direction, approximately the contrast between the first two predictors; least squares amplifies noise in that direction.

Answer (b)

The following simulation keeps the standardized design fixed. Each repetition generates one independent training/test response pair and reuses it for all three penalties. All predictions include the unpenalized training intercept.

The empirical coefficient summaries from the 200 repetitions are:

λ	Quantity	Mean	SD
0	$\hat{\beta}_1$	1.495589	1.543948
0	$\hat{\beta}_2$	1.509765	1.548588
0	$\hat{\beta}_1 + \hat{\beta}_2$	3.005354	0.095108
0	$\hat{\beta}_1 - \hat{\beta}_2$	-0.014175	3.091078
0.02	$\hat{\beta}_1$	1.486994	0.161513
0.02	$\hat{\beta}_2$	1.489176	0.165911
0.02	$\hat{\beta}_1 + \hat{\beta}_2$	2.976171	0.094165
0.02	$\hat{\beta}_1 - \hat{\beta}_2$	-0.002182	0.313622
0.2	$\hat{\beta}_1$	1.367908	0.045703
0.2	$\hat{\beta}_2$	1.368693	0.047337
0.2	$\hat{\beta}_1 + \hat{\beta}_2$	2.736601	0.086467
0.2	$\hat{\beta}_1 - \hat{\beta}_2$	-0.000785	0.034388

λ	Average training MSE	Average test MSE
0	0.964756	1.062452
0.02	0.974681	1.049262
0.2	1.076760	1.133484

Answer (c)

The true first two coefficients are both 1.5, with sum 3 and difference 0. OLS has empirical means close to these targets, but its individual coefficient SDs are about 1.54 and its difference SD is 3.091078. With $\lambda = 0.02$, the individual SDs fall to about 0.16 while the means remain close to 1.5. This small bias accompanies a large variance reduction, and average test MSE decreases from 1.062452 to 1.049262 in this simulation. With $\lambda = 0.2$, the means fall to about 1.368: the additional shrinkage introduces enough prediction bias to outweigh the further variance reduction, giving test MSE 1.133484. OLS has the lowest training MSE because it minimizes the unpenalized training residual sum of squares; that does not guarantee the lowest test error.

To understand why fitted values change much less than individual coefficients, write the contribution of the first two predictors as

$$X_1\hat{\beta}_1 + X_2\hat{\beta}_2 = \frac{X_1 + X_2}{2}(\hat{\beta}_1 + \hat{\beta}_2) + \frac{X_1 - X_2}{2}(\hat{\beta}_1 - \hat{\beta}_2).$$

Because $X_1 - X_2$ is small, even a large coefficient difference contributes little to prediction. Ridge strongly stabilizes this weak direction, while modest penalization barely changes the variability of the well-estimated sum (SD 0.095108 versus 0.094165). Thus a dramatic improvement in coefficient stability can yield only a modest improvement in prediction error. Independent test noise also remains present regardless of the penalty.

Question 3: What does ridge regression shrink?

Answer (a)

Expanding the squared residual norm gives

$$L_\lambda(\beta) = \frac{1}{2n} (\tilde{\mathbf{y}}^\top \tilde{\mathbf{y}} - 2\beta^\top \mathbf{X}^\top \tilde{\mathbf{y}} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta) + \frac{\lambda}{2} \beta^\top \beta.$$

Since $\mathbf{X}^\top \mathbf{X}$ is symmetric, differentiation with respect to the slope vector yields

$$\nabla L_\lambda(\beta) = -\frac{1}{n} \mathbf{X}^\top \tilde{\mathbf{y}} + \frac{1}{n} \mathbf{X}^\top \mathbf{X} \beta + \lambda \beta.$$

Setting the gradient equal to zero and multiplying by n gives

$$(\mathbf{X}^\top \mathbf{X} + n\lambda \mathbf{I}_p) \hat{\beta}_\lambda = \mathbf{X}^\top \tilde{\mathbf{y}}.$$

To establish uniqueness, consider the Hessian

$$\nabla^2 L_\lambda = \frac{1}{n} \mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p.$$

For every nonzero $\mathbf{a} \in \mathbb{R}^p$ and every $\lambda > 0$,

$$\mathbf{a}^\top \nabla^2 L_\lambda \mathbf{a} = \frac{1}{n} \|\mathbf{X}\mathbf{a}\|_2^2 + \lambda \|\mathbf{a}\|_2^2 > 0.$$

Thus the objective is strictly convex and the normal-equation matrix is positive definite and invertible. Its stationary point is the unique global minimizer:

$$\hat{\beta}_\lambda = (\mathbf{X}^\top \mathbf{X} + n\lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \tilde{\mathbf{y}}.$$

This argument does not require full column rank. Even if $\mathbf{X}\mathbf{a} = \mathbf{0}$ for a nonzero direction, as can occur with rank deficiency or $p > n$, the penalty contributes the strictly positive term $\lambda \|\mathbf{a}\|_2^2$. The intercept is fitted separately and remains unpenalized.

Answer (b)

The compact SVD has orthonormal columns in $\mathbf{U}_r$ and $\mathbf{V}_r$, and positive diagonal entries d_j in $\mathbf{D}_r$. Hence

$$\mathbf{X}^\top \mathbf{X} = \mathbf{V}_r \mathbf{D}_r^2 \mathbf{V}_r^\top, \quad \mathbf{X}^\top \tilde{\mathbf{y}} = \mathbf{V}_r \mathbf{D}_r \mathbf{U}_r^\top \tilde{\mathbf{y}}.$$

In particular,

$$(\mathbf{X}^\top \mathbf{X} + n\lambda \mathbf{I}_p) \mathbf{V}_r = \mathbf{V}_r (\mathbf{D}_r^2 + n\lambda \mathbf{I}_r).$$

Multiplying by the appropriate inverses therefore gives

$$\hat{\beta}_\lambda = \mathbf{V}_r (\mathbf{D}_r^2 + n\lambda \mathbf{I}_r)^{-1} \mathbf{D}_r \mathbf{U}_r^\top \tilde{\mathbf{y}}.$$

Any component orthogonal to the columns of $\mathbf{V}_r$ lies in the null space of $\mathbf{X}$. It cannot improve the residual fit, so the positive ridge penalty sets it to zero. Multiplying the slope estimate by $\mathbf{X}$ yields the centered fitted response:

$$\begin{aligned}\mathbf{X}\hat{\beta}_\lambda &= \mathbf{U}_r \mathbf{D}_r (\mathbf{D}_r^2 + n\lambda \mathbf{I}_r)^{-1} \mathbf{D}_r \mathbf{U}_r^\top \tilde{\mathbf{y}} \\ &= \sum_{j=1}^r \underbrace{\frac{d_j^2}{d_j^2 + n\lambda}}_{\rho_j(\lambda)} \mathbf{u}_j \mathbf{u}_j^\top \tilde{\mathbf{y}}.\end{aligned}$$

The least-squares fitted component in direction $\mathbf{u}_j$ is $\mathbf{u}_j \mathbf{u}_j^\top \tilde{\mathbf{y}}$. Ridge multiplies this component by $\rho_j(\lambda)$, a number between zero and one for $\lambda > 0$.

Here $n\lambda = 100(0.01) = 1$, so

$$\begin{aligned}\rho_1(0.01) &= \frac{10^2}{10^2 + 1} = \frac{100}{101} \approx 0.990099, \\ \rho_2(0.01) &= \frac{2^2}{2^2 + 1} = \frac{4}{5} = 0.8, \\ \rho_3(0.01) &= \frac{0.2^2}{0.2^2 + 1} = \frac{0.04}{1.04} = \frac{1}{26} \approx 0.038462.\end{aligned}$$

Because the predictor columns are centered, their left singular vectors are orthogonal to the constant-vector direction. The fitted intercept retains that direction with multiplier one. Consequently,

$$\text{df}_{\text{eff}}(0.01) = 1 + \frac{100}{101} + \frac{4}{5} + \frac{1}{26} = \frac{37139}{13130} \approx \boxed{2.828561}.$$

This is an effective response-sensitivity measure, not a count of nonzero coefficients.

Answer (c)

The direction $\mathbf{u}_3$, corresponding to $d_3 = 0.2$, receives the strongest shrinkage: ridge retains only about 3.85% of its least-squares fitted component. Smaller singular values give smaller values of $\rho_j(\lambda)$ at a fixed positive penalty.

Under the lecture's fixed-design noise model, let $\theta_j = \mathbf{u}_j^\top \mu$ and let the error variance be σ^2 . The ridge fitted coordinate is

$$\hat{\theta}_{j,\lambda} = \rho_j(\lambda) \mathbf{u}_j^\top \tilde{\mathbf{y}},$$

so

$$\begin{aligned}\text{Var}(\hat{\theta}_{j,\lambda}) &= \rho_j(\lambda)^2 \sigma^2, \\ \text{Bias}(\hat{\theta}_{j,\lambda}) &= (\rho_j(\lambda) - 1) \theta_j.\end{aligned}$$

The third coordinate's variance falls from σ^2 to $\sigma^2/676$, substantially reducing prediction variance. If the true signal θ_3 is large, however, the bias $-(25/26)\theta_3$ can produce enough squared bias to outweigh that benefit.

Question 4: Ridge regression and optimization

Answer (a)

For a symmetric positive-definite curvature matrix, the spectral condition number is the ratio of its largest to smallest eigenvalue. Thus

$$\kappa(\mathbf{A}_0) = \frac{9}{0.01} = 900.$$

Adding $\lambda \mathbf{I}_p$ shifts every eigenvalue by λ without changing the eigendirections. At $\lambda = 0.09$, the ridge eigenvalues are

$$9 + 0.09 = 9.09, \quad 0.01 + 0.09 = 0.10,$$

and therefore

$$\kappa(\mathbf{A}_\lambda) = \frac{9.09}{0.10} = 90.9.$$

Ridge makes the objective's elliptical contours less elongated. The initially flat direction becomes ten times as curved, while the steep direction changes only slightly, so the curvature is more balanced.

Answer (b)

The gradient of the stated ridge objective is

$$\nabla L_\lambda(\beta) = \mathbf{A}_\lambda \beta - \frac{1}{n} \mathbf{X}^\top \tilde{\mathbf{y}}.$$

At the ridge minimizer the gradient vanishes, so

$$\mathbf{A}_\lambda \hat{\beta}_\lambda = \frac{1}{n} \mathbf{X}^\top \tilde{\mathbf{y}}, \quad \nabla L_\lambda(\beta^{(k)}) = \mathbf{A}_\lambda \mathbf{e}^{(k)}.$$

Subtracting the minimizer from both sides of the gradient-descent update gives

$$\begin{aligned} \mathbf{e}^{(k+1)} &= \beta^{(k)} - \eta \mathbf{A}_\lambda \mathbf{e}^{(k)} - \hat{\beta}_\lambda \\ &= (\mathbf{I}_p - \eta \mathbf{A}_\lambda) \mathbf{e}^{(k)}. \end{aligned}$$

Let $\mathbf{v}$ be a unit eigenvector of $\mathbf{A}_0$ with eigenvalue a . Then $\mathbf{A}_\lambda \mathbf{v} = (a + \lambda)\mathbf{v}$. By symmetry of $\mathbf{A}_\lambda$, the scalar error coordinate satisfies

$$\mathbf{v}^\top \mathbf{e}^{(k+1)} = [1 - \eta(a + \lambda)] \mathbf{v}^\top \mathbf{e}^{(k)}.$$

Thus each iteration multiplies that coordinate by $1 - \eta(a + \lambda)$. Strict contraction requires its absolute value to be less than one:

$$\begin{aligned} |1 - \eta(a + \lambda)| < 1 &\iff -1 < 1 - \eta(a + \lambda) < 1 \\ &\iff 0 < \eta(a + \lambda) < 2 \\ &\iff 0 < \eta < \frac{2}{a + \lambda}. \end{aligned}$$

Both ridge eigenvalues are positive. Intersecting their allowable intervals yields

$$0 < \eta < \min \left\{ \frac{2}{9.09}, \frac{2}{0.10} \right\} = \frac{2}{9.09}.$$

Equivalently, the full interval is

$$\boxed{\eta \in \left(0, \frac{2}{M}\right) = \left(0, \frac{200}{909}\right) \approx (0, 0.220022)}.$$

The endpoints are excluded: at $\eta = 0$ no error decreases, and at $\eta = 2/M$ the largest-curvature coordinate has multiplier -1 and does not contract.

For $\eta = 1/M = 1/9.09 \approx 0.110011$, the two multipliers are

$$\begin{aligned} a = 9 : \quad 1 - \eta(a + \lambda) &= 1 - \frac{9.09}{9.09} = 0, \\ a = 0.01 : \quad 1 - \eta(a + \lambda) &= 1 - \frac{0.10}{9.09} = \frac{899}{909} \approx 0.988999. \end{aligned}$$

The high-curvature direction converges faster: its optimization error is eliminated in one step in exact arithmetic. The low-curvature error retains about 98.90% of its value each step, so it decays slowly and controls the eventual convergence rate.

For $\eta = 2.1/M = 2.1/9.09 \approx 0.231023$, the step exceeds $2/M$. In the largest-curvature direction,

$$1 - \eta M = 1 - 2.1 = -1.1, \quad \mathbf{v}_M^\top \mathbf{e}^{(k)} = (-1.1)^k \mathbf{v}_M^\top \mathbf{e}^{(0)}.$$

Under the stated nonzero-initial-component assumption, this error alternates sign and grows in magnitude by a factor of 1.1 at every step. Gradient descent therefore diverges. Although the low-curvature multiplier is

$$1 - \frac{2.1(0.10)}{9.09} = 1 - \frac{0.21}{9.09} = \frac{296}{303} \approx 0.976898,$$

contraction in that direction cannot compensate for growth in the orthogonal largest-curvature direction.

Answer (c)

The penalty λ changes the statistical estimator: stronger shrinkage can reduce variance but increase prediction bias. Better conditioning alone therefore does not imply better predictions. Choose λ using a training-only prediction criterion such as cross-validation.

For fixed λ , the optimization algorithm and step size control how efficiently and reliably the same ridge minimizer is reached. A convergent gradient-descent run and a linear-system solve target the same estimate. The step size must respect the curvature bound; it does not choose the statistical bias-variance tradeoff.

Question 5: Selecting the ridge penalty for real-estate prediction

Answer (a)

The files are matched by `row_id`; the resulting split has 332 training observations and 82 test observations. The six quantitative predictors are `date`, `age`, `distance`, `stores`, `latitude`, and `longitude`. The grid contains 122 candidates: zero and the 121 specified positive penalties. Within each fold, predictor means, predictor scales (using divisor equal to the fitting sample size), and the response mean are estimated from the other nine folds only. Validation predictions reuse those quantities.

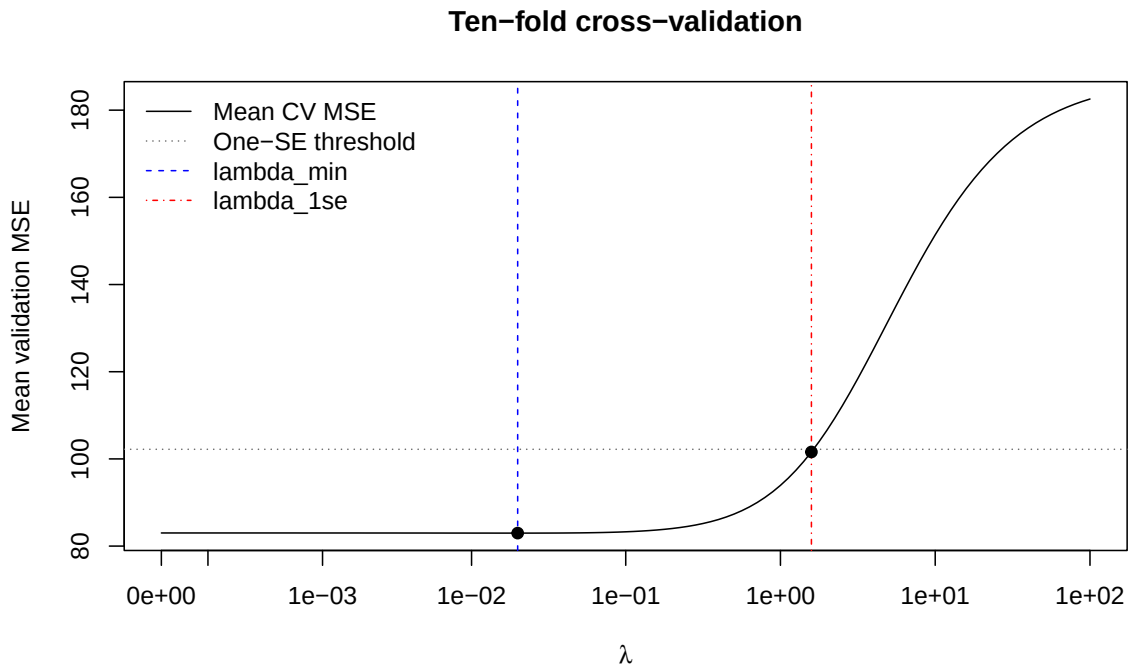


Figure 1: Mean validation MSE over the exact penalty grid. The shifted logarithmic horizontal axis includes OLS at zero; the horizontal line is the one-SE threshold.

The mean validation MSE weights the ten fold MSEs equally. Its SE is their sample standard deviation divided by $\sqrt{10}$. The complete grid summaries are retained in `cv_results`. The selected candidates have:

Selection	λ	Mean validation MSE	SE
Minimum CV error	0.01995262	82.97765	19.22991
One-SE rule	1.58489319	101.59553	19.97678

The one-SE cutoff is $82.97765 + 19.22991 \approx 102.20756$, and $\lambda_{1se} = 1.58489319$ is the largest grid candidate whose mean CV MSE does not exceed that cutoff. The cutoff uses the SE at the minimum.

Answer (b)

For GCV, preprocessing is estimated once from all 332 training observations. If d_j are the singular values of this standardized training matrix, the calculation uses

$$\text{df}_{\text{eff}}(\lambda) = 1 + \sum_{j=1}^6 \frac{d_j^2}{d_j^2 + 332\lambda}, \quad \text{GCV}(\lambda) = \frac{\text{MSE}_{\text{train}}(\lambda)}{\{1 - \text{df}_{\text{eff}}(\lambda)/332\}^2}.$$

The leading one counts the unpenalized intercept.

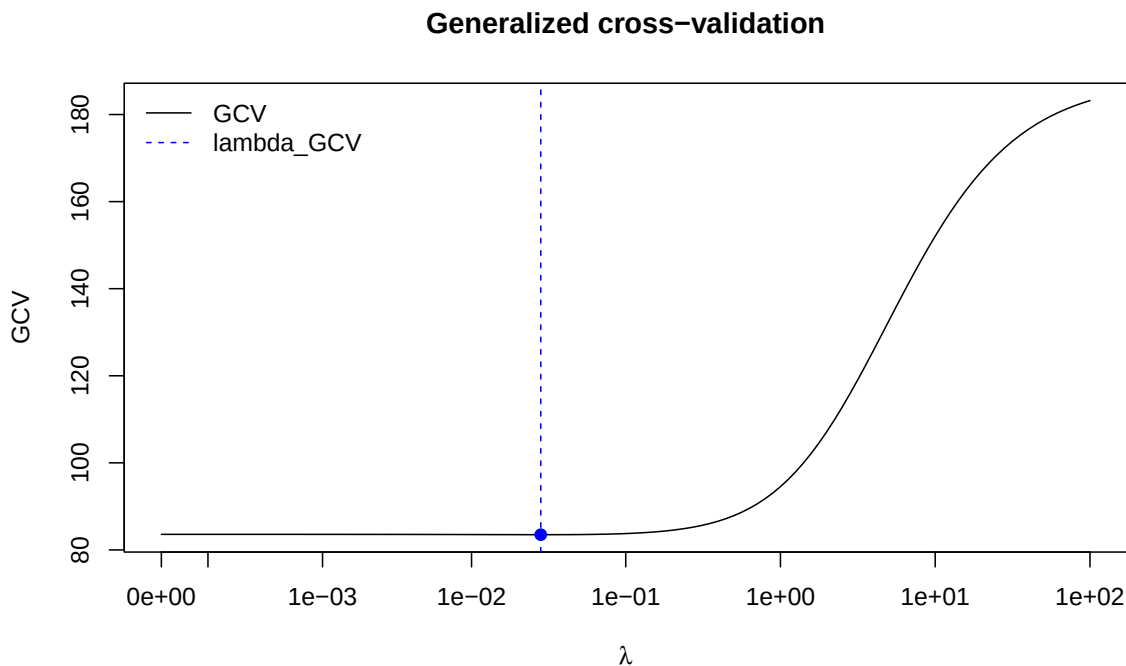


Figure 2: Training-only GCV over the same penalty grid, with the selected penalty marked. The shifted logarithmic horizontal axis includes zero.

The minimum occurs at $\lambda_{\text{GCV}} = 0.02818383$, with GCV **83.50191**, training MSE **80.17125**, and effective degrees of freedom **6.688657**.

Answer (c)

All three penalties are fixed before the following code accesses test responses. The four models are refitted on the full training set, and test predictors are transformed using the full-training means and scales. Predictions include the training response mean.

Model	λ	Training MSE	Test MSE	Effective df
OLS	0	80.093917	66.322464	7.000000
Ridge, $\lambda_{\min}$	0.01995262	80.135429	65.999575	6.773213

Model	λ	Training MSE	Test MSE	Effective df
Ridge, λ_{1se}	1.58489319	100.373559	81.199810	3.032870
Ridge, λ_{GCV}	0.02818383	80.171254	65.902765	6.688657

OLS minimizes training MSE. The CV-minimum and GCV fits apply modest shrinkage and have slightly lower observed test MSEs; these small differences do not establish reliable superiority from one split. The one-SE fit is more strongly regularized and has higher training and test errors, consistent with greater shrinkage bias. The one-SE heuristic does not guarantee the lowest test error.

Although the GCV fit has the smallest observed test MSE, selecting it on that basis would turn the test set into tuning data and make the reported minimum optimistically selected. Choose the procedure using training-only evidence and use the test set to evaluate the fixed choices. Further tuning would require new independent evaluation data.

explain-to-me skill

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name: explain-to-me
description: Help me understand what a homework question is asking without
  ↪ solving it. Use only when I explicitly ask you to use the explain-to-me
  ↪ skill.
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# Intent

Help me understand a homework question so I can complete the work myself.

# Instructions

- Read the full question before responding.
- Explain the main goal of the question in simple language.
- Break down what work the question is asking me to do.
- Clarify any notation, terminology, or statistical concepts that may be
  ↪ confusing.
- Give me a reasonable first step or small hint if helpful.
- Do not perform the calculations, write the solution code, complete
  ↪ derivations, or give me the final answer.
- Keep the explanation focused and conversational.
```